5 Minor/Department/Interdisciplinary Specialization for batches undergoing transition plan

Programme Structure

S.N.	Course Type	Credits
1	Core	8
2	Electives	12
	Total for award of Minor/Interdisciplinary/Department Specialization	20

Note: Project credit should be less than or equal to 6 out of 12 elective credits except for Engineering Innovation and Entrepreneurship

5.1 Artificial Intelligence (CS/EE/ME/BB)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	Machine Learning 1	3-0-0	3
2	Artificial Intelligence 2[#]	3-0-0	3
3	Real-time Autonomous Systems	2-0-0	2

Note: [#]ME, EE and BB students will take Artificial Intelligence 1 as a compulsory course and Artificial Intelligence 2 as an elective course.

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Machine Learning II	3-0-0	3
2	Algorithms for Big Data	2-0-0	2
3	Computer Vision	3-0-0	3
4	Stream Analytics	3-0-0	3
5	Machine Learning with Big Data	3-0-0	3
6	Digital Image Analysis	3-0-0	3
7	Edge and Fog Computing	3-0-0	3
8	Natural Language Processing	3-0-0	3
9	GPU Programming	3-0-0	3
10	Data Visualization	3-0-0	3
11	Introduction to Augmented Reality and Virtual Reality	3-0-0	3
12	Dependable AI	3-0-0	3
13	Resource Constrained AI	3-0-0	3
14	Special Topics in Artificial Intelligence - I	3-0-0	3
15	Special Topics in Artificial Intelligence - II	3-0-0	3
16	Special Topics in Artificial Intelligence - III	3-0-0	3
17	Graph Theory and Applications	3-0-0	3
18	Social Network Analysis	3-0-0	3
19	Computational Optimization	3-0-0	3
20	Computer Graphics	3-0-0	3
21	Advanced Computer Graphics	3-0-0	3
22	Neuromorphic Computing and Design	3-0-0	3
23	Bio-image Computing	3-0-0	3
24	Project	0-0-12	6

5.2 VLSI (CS/EE)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	Digital VLSI Design	3-0-0	3
2	Analog and Interfacing Circuits	3-0-0	3
3	Integrated Circuits Design Lab	1-0-2	2

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Flexible and Printed Electronics	3-0-0	3
2	RF IC Design	3-0-0	3
3	Active Filter Design	3-0-0	3
4	Image Sensor Design and Applications	3-0-0	3
5	Physics of Transistors	3-0-0	3
6	IC Technology	3-0-0	3
7	Neuromorphic Computing and Design	3-0-0	3
8	Project		6

5.3 Internet of Things (CS/EE)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	Introduction to Internet of Things	1-0-0	1
2	Sensors and Measurement	3-0-0	3
3	Data communication and Networking	3-0-0	3
4	Sensors and IoT Lab	1-0-0	1

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Digital and Systems-on-Chip Design	3-0-0	3
2	Active Filter Design	3-0-0	3
3	Embedded Systems Design	3-0-0	3
4	Image Sensor Design and Applications	3-0-0	3
5	Antenna Engineering	3-0-0	3
6	Wireless Communication	3-0-0	3
7	Advanced Digital Signal Processing	3-0-0	3
8	Analog and Interfacing Circuits	3-0-0	3
9	Machine Learning - I	3-0-0	3
10	Fog and Edge Computing	3-0-0	3
11	Resource Constrained AI	3-0-0	3
12	Project		6

5.4 Cyber Physical Systems (CS/EE/ME/BB)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	Introduction to Cyber physical Systems	3-0-0	3
2	Embedded System Design	3-0-0	3
3	Embedded System Design Laboratory	0-0-2	1
4	Introduction to Modelling (Fractal offered by Dept. Of Mathematics)	1-0-0	1

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Computer Vision	3-0-0	3
2	Advanced Digital Signal Processing	3-0-0	3
3	Vehicular Ad-Hoc Networks(VANETs)	3-0-0	3
4	Edge & Fog Computing	3-0-0	3
5	Robotics : Fractal 1 Robot Modelling	1-0-2	2
6	Robotics : Fractal 2 Control	1-0-0	1
7	Robotics : Fractal 3 Motion Planning and Programming	1-0-0	1
8	Machine Learning I	3-0-0	3
9	Digital and Systems-on-Chips Design	3-0-0	3
10	Introduction to Virtual Reality	3-0-0	3
11	Smart Grid	3-0-0	3
12	Adaptive Signal processing	3-0-0	3
13	Computer Architecture	3-0-0	3
14	Digital Twin	3-0-0	3
15	Advanced Control System	3-0-0	3
16	Security of CPS	2-0-0	2
17	Machine Learning II	3-0-0	3
18	Artificial Intelligence I	3-0-0	3
19	Artificial Intelligence II	3-0-0	3
20	Smart Manufacturing	1-0-2	2
21	Resource Constrained AI	3-0-0	3
22	Machine Learning II	3-0-0	3
23	Industry 4.0 Applications in Manufacturing Systems	1-0-0	1
24	Optimization	1-0-0	1
25	Project-1		6

5.5 Communications and Signal Processing (EE)

A. Compulsory Courses

S.N.	Course Type	L-T-P	Credits
1	Estimation Theory	1-0-0	1
2	Multiple Parameter Estimation	1-0-0	1
3	Wireless Communications	3-0-0	3
4	Data Communication and Networking	3-0-0	3

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Advanced Digital Signal Processing	3-0-0	3
2	MIMO Wireless Communications	3-0-0	3
3	Antenna Engineering	3-0-0	3
4	Network Information Theory	3-0-0	3
5	Statistical Decision Theory	2-0-0	2
6	Multi-rate Digital Signal Processing	2-0-0	2
7	Digital Image Processing and Applications	3-0-0	3
8	Digital Video Processing	3-0-0	3
9	RF IC Design	3-0-0	3
10	Adaptive Signal Processing	3-0-0	3
11	Optical Fiber Communications	3-0-0	3
12	Image sensor Design and Application	3-0-0	3
13	Applied Optimization for Wireless Communication	3-0-0	3
14	Machine Learning for Communications	3-0-0	3
15	Machine Learning 1	3-0-0	3
16	Physical Layer Security	3-0-0	3
17	Compressive Sensing	3-0-0	3
18	Bio-Image Computing	3-0-0	3
19	Selected Topics in Signal Processing I	1-0-0	1
20	Selected Topics in Signal Processing II	2-0-0	2
21	Selected Topics in Signal Processing III	3-0-0	3
22	Multimedia Communications	3-0-0	3
23	Free Space Optical Communications	3-0-0	3
24	GNSS Signal Processing	3-0-0	3
25	Backhaul Networks for Wireless Systems	3-0-0	3
26	Selected Topics in Communication I	1-0-0	1
27	Selected Topics in Communication II	2-0-0	2
28	Selected Topics in Communication III	3-0-0	3
29	Neuromorphic computing and design	3-0-0	3
30	Machine Learning 2	3-0-0	3
31	Real Time Communications	3-0-0	3
32	Millimeter Wave Technology	3-0-0	3
33	Digital Image and Video Processing Lab	1-0-2	2
34	RF IC Design Lab	1-0-2	2
35	Network Information Theory	3-0-0	3
36	Wavelets	3-0-0	3
37	Computational Imaging	3-0-0	3
38	Speech and Audio Processing	3-0-0	3
39	Data Compression	3-0-0	3
40	Machine Learning for Communication Lab	1-0-2	2

5.6 Design (ME)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	EE: Introduction to Sensors (Fractal Course)	1-0-0	1
2	MEL7XX0: Geometric Modeling & CAD	2-0-2	3
3	MA: Optimization (Fractal course)	1-0-0	1
4	MEL7XX0: Advanced Mechanics of Solids	3-0-0	3

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	MEL7XX0: Robotic	3-0-2	4
2	MEL7XX0: Theory of Vibrations	3-0-0	3
3	MEL7XX0: Mechanical Vibrations	3-0-0	3
4	MEL7XX0: Helicopter Dynamics	3-0-0	3
5	MEL7XX0: Modeling and Control of Mechanical Systems	3-0-0	3
6	MEL7XX0: Multibody Dynamics	3-0-0	3
7	MEL7XX0: Finite Element Methods in Engineering	3-0-0	3
8	MEL7XX0: Rotor Dynamics	3-0-0	3
9	AVL7XX0: Autonomous Mobile Robots	3-0-2	4
10	BBXXX: Biomechanics	3-0-0	3
11	MEL7XX0: MEMS and Microsystems Technology	3-0-2	4
12	MEL7XX0: Applied Ergonomics	3-0-0	3
13	MEL7XX0: Mechatronic Systems Design	3-0-0	3
14	MEL7XX0: Additive Manufacturing	3-0-0	3
15	MEL7XX0: Design for X	3-0-0	3
16	MEL7XX0: Computer Aided Inspection and Quality Control	3-0-0	3
17	MEL7XX0: Microfluidics Technology	3-0-0	3
18	BBXXX: Medical Devices and Implants	3-0-0	3
19	MTXXX: Light Metals and Alloys	3-0-0	3
20	MTXXX: Corrosion Engineering	3-0-0	3
21	MAXXX: Statistics I	1-0-0	3
22	MAXXX: Statistics II	2-0-0	3
23	MAXXX: Reliability Theory	3-0-0	3
24	CSXXX: Artificial Intelligence I	3-0-0	3
25	CSXXX: Machine Learning I	3-0-0	3
26	MEL7XX0: Probabilistic Methods in Engineering Design	3-0-0	3
27	PHL7XX0: Energy Harvesting Technologies	3-0-0	3

5.7 Smart Manufacturing (ME)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	EE: Introduction to Sensors (Fractal Course)	1-0-0	1
2	CS: Introduction to Machine Learning (Fractal course)	1-0-0	1
3	MEP7XX0: Smart Manufacturing	1-0-2	2
4	MEL7XX0: Materials & Manufacturing Processes	2-0-1	3

5	MEL7XX0: Industry 4.0 Applications in Manufacturing Systems	1-0-0	1

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	MEL7XX0: Geometric Modeling & CAD	2-0-1	3
2	MA: Optimization (Fractal course)	1-0-0	1
3	MEL7XX0: Robotic	3-0-2	4
4	MEL7XX0: Computer Aided Manufacturing	3-0-2	4
5	MEL7XX0: Mechanical Metallurgy	3-0-0	3
6	MEL7XX0: Theory of Arc Welding	3-0-0	3
7	MEL7XX0: Welding Metallurgy	3-0-0	3
8	MEL7XX0: MEMS and Microsystems Technology	3-0-2	4
9	MEL7XX0: Precision and Micro-Manufacturing	3-0-0	3
10	MEL7XX0: Applied Ergonomics	3-0-0	3
11	MEL7XX0: Mechatronic Systems Design	3-0-0	3
12	MEL7XX0: Additive Manufacturing	3-0-0	3
13	MEL7XX0: Design for X	3-0-0	3
14	MEL7XX0: Computer Aided Inspection and Quality Control	3-0-0	3
15	MEL7XX0: Welding: Metallurgy, Defects and Testing	3-0-0	3
16	MEL7XX0: Non-Conventional Welding Processes	3-0-0	3
17	MEL7XX0: Welding inspection and Quality Control	3-0-0	3
18	MEL7XX0: Microfluidics Technology	3-0-0	3
19	BBXXX: Medical Devices and Implants	3-0-0	3
20	MTXXX: Solidification Processing	3-0-0	3
21	MTXXX: Light Metals and Alloys	3-0-0	3
22	MTXXX: Powder Metallurgy	3-0-0	3
23	MTXXX: Thermo-Mechanical Processing	3-0-0	3
24	MTXXX: Near-Net Forming	3-0-0	3
25	MTXXX: Plastic Deformation and Microstructure Evaluation	3-0-0	3
26	MTXXX: Corrosion Engineering	3-0-0	3
27	MAXXX: Statistics I	3-0-0	3
28	MAXXX: Statistics II	3-0-0	3
29	MAXXX: Reliability Theory	3-0-0	3
30	CSXXX: Artificial Intelligence I	3-0-0	3
31	CSXXX: Machine Learning I	3-0-0	3
32	EEXXX: Introduction to CPS	3-0-0	3
33	MEL7XX0: Probabilistic Methods in Engineering Design	3-0-0	3
34	PHL7XX0: Nanoscience and Nanotechnology	3-0-0	3
35	PHL7XX0: Introduction to Material Characterization	3-0-0	3
36	PHL7XX0: Physics of Solar Cells	3-0-0	3
37	PHL7XX0: Laser Technologies	3-0-0	3
38	PHL7XX0: Computational Materials Science	3-0-0	3
39	PHL7XX0: Statistical Methods for Data Analysis	3-0-0	3
40	PHL7XX0: Energy Harvesting Technologies	3-0-0	3

5.8 Thermofluids Engineering (ME)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	MEL7XX0: Applied Heat and Mass Transfer	3-0-0	3
2	MEL7XX0; Incompressible Fluid Flow	3-0-0	3
3	MEL7XX1: Multi Phase Systems and Chemical Reactions	1-0-0	1
4	MEL7XX2: Kinetic Theory	1-0-0	1

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	MEL7XX0: Experimental Techniques in Thermofluids	3-0-0	3
2	MA: Matrix Computations	1-0-0	1
3	MA: Numerical Methods in ODE	1-0-0	1
4	MA: Numerical Methods for PDE	1-0-0	1
5	MEL7XX0: Conduction and Radiation	3-1-0	4
6	MEL7XX0: Convective Heat and Mass Transfer	3-1-0	4
7	MELXXX0: Combustion Technology	3-0-0	3
8	MEL7XX0: Turbulent Flows	3-0-2	4
9	MEL7XX0: Jet and Rocket Propulsion	3-0-0	3
10	MEL7XX0: Computational Fluid Dynamics and Heat Transfer	3-1-0	4
11	MEL7XX0: Two Phase Flow and Heat Transfer	3-1-0	4
12	MEL7XX0: Design of Heat Exchange Equipment	3-1-0	4
13	MEL7XX0: Compressible Fluid Flows	3-0-2	4
14	MEL7XX0: Optical and Laser Based Measurements in Thermofluids	3-0-0	3
15	MEL7XX0: Fluid dynamics and Acoustics of Turbulent Jets	3-0-0	3
16	MEL7XX0: Heating, Ventilation and Air-Conditioning	3-0-0	3
17	MEL7XX0: Microfluidics Technology	3-0-0	3
18	MEL7XX0: Fluid Flow in Biological Processes	3-0-0	3
19	MEL7XX0: Nanosensors	3-0-0	3
20	MEL7XX0: Water Energy Nexus	3-0-0	3
21	MEL7XX0: Soil and Water Conservation Engineering	3-0-0	3
22	MEL7XX0: Probabilistic Methods in Engineering Design	3-0-2	4

5.9 Smart Healthcare (CS/EE/ME/BB)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	Disease, processes, diagnostics & therapeutics	3-0-2	4
2	Systems Biology	3-0-2	4

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Bioimaging	3-0-0	3
2	Artificial Intelligence for healthcare	3-0-0	3
3	Biosensors	3-0-0	3
4	Multiomics technologies	3-0-0	3
5	Biomaterials	3-0-0	3
6	Immunotechnology	3-0-0	3
7	Quantitative Physiology lab	0-0-2	1
8	Statistics and Data analysis	3-0-2	4
9	Principles of drug discovery	3-0-0	3
10	Principles of Nuclear Magnetic Resonance	3-0-0	3
11	Theory & Applications of Spectroscopy	3-0-0	3
12	Project work	0-0-8	4

5.10 Management (CS/EE/ME/BB)

A. Core Courses

S.N.	Course Type	L-T-P	Credits
1	Marketing Management-I	2-0-0	2
2	Managing Human Capital	2-0-0	2
3	Management Accounting	2-0-0	2
4	Operations Management	2-0-0	2

B. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	Macroeconomic Environment of Business	2-0-0	2
2	Corporate Finance-I	2-0-0	2
3	Statistics for Management	2-0-0	2
4	Decision Sciences-I	2-0-0	2
5	Managerial Economics	2-0-0	2
6	Managing Digital Enterprises	2-0-0	2
7	Legal Aspects of Business	2-0-0	2
8	Strategy Formulation	2-0-0	2
9	Econometrics	2-0-0	2
10	Managing New Ventures	2-0-0	2
11	Managing Innovation	2-0-0	2
12	Strategy Execution	1-0-0	1
13	Corporate Finance-II	1-0-0	1
14	Business Research Methods	1-0-0	1
15	Global Economy	1-0-0	1
16	Marketing Management-II	1-0-0	1
17	Decision Sciences-II	1-0-0	1

Maximum 4 credits can be earned through workshop

Note:

- As this plan is only for transition, the course on Principles of Management and Economics is not included here. Existing students have already done these courses as a life skill course. Generally Principles of Management and Economics is a core component.

5.11 Entrepreneurship (CS/EE/ME/BB)

A. Core Courses

S. N.	Course Name	L-T-P	Credits
1	Starting up New Venture	2-0-0	2
2	Ideation & Prototyping	2-0-0	2
3	New Venture Financing	2-0-0	2
4	Business Plan Phase-I: Problem Selection, Concept Development and Commercial Viability Assessment (Ideation, Product Concept Development, Product Viability, Costing and Pricing & Market Traction) (Project-based course)	0-0-8	4
5	Business Plan Phase–II: Prototype Development and Business Model Development (Prototype Development [Development of Minimum Viable Product (MVP) for Customer Discovery and Product-Market Fit], Business Model Development, IP strategy formulation) (Project-based course)	0-0-16	8
6	IPR & Legal Issues for Startups	1-0-0	1
7	Systems Engineering and Project Management	1-0-0	1

Note:

- 'Minor in Entrepreneurship' will be mostly project-based involving hands-on exercise in the classroom and experiential learning in the lab/field. While students will earn 08 credits from theory courses, 12 credits can be garnered from project work spread across two semesters (one-year)
-
- As project ideas are expected to be technology-driven, faculty members from Engineering/Technology departments will contribute to developing and executing the core-courses as well as business plans along with the faculty from SME. The endeavour will be made to involve relevant entrepreneurs in the delivery of these courses.
-
- The students keen to pursue business plans that have a market potential may get the opportunity to enter MTech in Entrepreneurship programme (it is yet to be planned).
-
- Only motivated students (or students group) will be admitted for the 'Minor in Entrepreneurship' based on initial scrutiny by a committee. Interested students will have to make a presentation demonstrating their interest and commitment for minor in entrepreneurship.

5.12 Robotics (CS/EE/ME/BB)

B. Core Courses

S.N.	Course Type	L-T-P	Credits
1	RMXX Introduction to Robotics	3-0-0	3
2	RMXX Autonomous Systems	3-0-0	3
3	RMXX Experimental Robotics	0-1-2	2

C. Elective Courses

S.N.	Course Type	L-T-P	Credits
1	RM7XX Advanced Robotic Manipulation	3-0-0	3
2	RM7XX Mobile Robots	3-0-0	3
3	RM7XX Aerial Robots	2-0-2	3
4	RM7XX Robot Operating System	3-0-0	3
5	RM7XX Selected Topics on Robotics	3-0-0	3
6	RM7XX Planning and Decision Making for Robots	3-0-0	3
7	AI3XX Artificial Intelligence (Non AI)/ AI4XX Advanced Artificial Intelligence (AI)	3-0-0	3
8	AI3XX Deep learning (Non AI)/ AI4XX Advanced Machine Learning (AI)	3-0-0	3
9	AI3XX Optimization (Non AI)	3-0-0	3
10	EE4XX Nonlinear Systems and Adaptive Control	3-0-0	3
11	EE7XX Networked Dynamical Systems and Control	3-0-0	3
12	EE7XX Advanced Control System	3-0-0	3
13	CS7XX Computer Vision	3-0-0	3
14	CS7XX Introduction to AR VR	3-0-0	3
15	EE7XX Introduction to Haptics	3-0-0	3
16	RM7XX Tele-Robotics	3-0-0	3
17	EE7XX Wireless Communication (CS/EE)	3-0-0	3
18	EE7XX UAV Assisted Wireless Networks (CS/EE)	3-0-0	3
19	Social Robotics	3-0-0	3
20	SH7XX Introduction to Medical Robotics	3-0-0	3
21	EE7XX Introduction to Cyber Physical Systems	3-0-0	3
22	AI4XX Introduction to Industry 4.0	3-0-0	3
23	ME7XX Smart Manufacturing	1-0-2	2
24	RM6XX Robotics Project (3 to 6 Credits)	0-0-12 (Max)	6 (Max)

Note:

1. Students should take at least total two courses from two baskets out of Robotics foundation, Learning and Control or Perception and Communication.
2. Students of transition batch can take
    - Artificial Intelligence 1(ME/EE/BB)/Artificial Intelligence 2 (CS) in lieu of courses under Sr. No. 7.
    - Machine Learning 1 in lieu of courses under Sr. No. 8.

5.13 Engineering Innovation (CS/EE/ME/BB)

Department Area Specialization in Engineering Innovation is aimed at making students Industry Ready by offering five months Practical Project in an Industry or R & D organization. It focuses on use of technology knowledge to create products and processes which have usability value. Objective of this specialisation is to recognise such creative engineering endeavour. This activity is not expected to aim only publication of papers. It must happen in collaboration with IIT J faculty member, and expected to have an industrial or User agency as partner. Each project carried out by students will have both a company supervisor and an institute supervisor. Students get real-world experience working for an industry or R & D organization on a project that complements their academic experience. Projects are aimed to enable them for meaningful careers and provide them with an opportunity to engage with industry before graduation.

In Engineering Innovation as a part of the curriculum from the beginning of the 8th Semester. A student would be required to complete a minimum of five months in industries or R & D organization or Institute for Department Area Specialization in Engineering Innovation.